

NAFIZ IMTIAZ KHAN

Davis, California, United States

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Career Objective

I am a PhD candidate in Computer Science at UC Davis, advised by Professor Vladimir Filkov. My research focuses on open-source software sustainability, health informatics, and applied data science. I specialize in fine-tuning deep neural networks, large language models, building agentic retrieval-augmented generation (RAG) systems, and developing task-specific AI/ML pipelines.

Education

University of California, Davis

PhD in Computer Science | MSc in Computer Science (2025)

Master's degree conferred July 2025, CGPA: 3.91/4.00

September 2023 - Present

California, United States

Technical Skills

Programming Languages: Python, C/C++, Java, JavaScript, PL/SQL, Typescript, HTML / CSS, Bash

Frameworks/Libraries : PyTorch, Keras, Tensorflow, LangChain, Scikit-learn, Pandas, NumPy, Matplotlib, BeautifulSoup, Pyvis, Node, React

Tools/Platform: Ollama, Flask, Git, AWS Bedrock, Google Colab, GCP, Docker, Kubernetes, Grafana, Selenium, Tableau, Microsoft Azure, JIRA

Database: ChromaDB, FAISS, MongoDB, FireBase, MySQL, Oracle, SQLite

Professional Experience

Human Frontier Collective Specialist – GenAI (Internship)

Aug 2025 – Sep 2025

Scale AI, Inc.

Remote – California, United States

- Designed and critiqued complex, domain-specific problem sets to rigorously test cutting-edge generative AI models, identifying limitations and refining performance.
- Collaborated with Scale researchers in interdisciplinary sessions to analyze model behavior, shape AI research directions, and explore experimental applications.

Graduate Student Researcher

September 2023 - Present

University of California, Davis

California, United States

- Developed **AngioVision** (<https://github.com/Nafiz43/AngioVision>), an LLM-powered angiography report generation and validation pipeline; fine-tuned a framework integrating a **CLIP** alignment module, embedding, and decoder layers, achieving a **10% F1 gain** over the prior SOTA, **Claude 4.6**.
- Led the end-to-end development of **RepoWise** (<https://repowise.github.io/RepoWise-website/>), a domain-aware RAG system for OSS governance analytics; demonstrated 84% accuracy on factual queries and superior performance over general-purpose LLMs (GPT-4o, Claude 4.5 Sonnet) in both factual and non-factual repository tasks.
- Spearheaded the development of the **OSSPREY** (<https://oss-prey.github.io/OSSPREY-Website/>) tool to assess the health of open-source projects, as measured by real-time analytics generation with an average latency of 3.5s and 124MB memory usage per project, by integrating **Vue.js** (frontend), **TensorFlow** (deep learning), and **LangChain & Ollama** (LLM inference pipeline design)
- Developed **ReACT-GPT** (<https://github.com/Nafiz43/ReACT-GPT>), an LLM-and-RAG-powered framework for extracting actionable insights from scientific articles, achieving a 75% BERTScore, by integrating **LangChain** (orchestration), **ChromaDB** (retrieval), and **Ollama** (lightweight LLM inference)
- Developed **EvidenceBot** (<https://nafiz43.github.io/EvidenceBot/>), a privacy-preserving **RAG** tool for querying and extracting insights from a diverse set of documents, built using **Streamlit** and powered by the **ReACT-GPT** framework
- Developed **PCL-Fetcher** (<https://nafiz43.github.io/PCL-Fetcher/>), an LLM-powered pipeline that automates procedural case log documentation from radiology reports, achieving an 86% F1Score—exceeding the crosswalk benchmark by over 13%
- Developed **MammoGen-RAG** (<https://github.com/Nafiz43/MammoGen-RAG>), a multi-modal RAG pipeline that leverages Vision-Language Models (VLMs) to automate mammography report generation, achieving a 90% F1score through integrated visual and textual reasoning
- Enhanced the performance of foundational language models in **LLM-as-Developers** (<https://github.com/zeerakb1/llm-developers-evaluation>) project by 10% (BERTScore) on answering StackOverflow questions through strategic fine-tuning techniques (**PEFT** and **QLoRA**)
- Developed and optimized **Transformer** and **LSTM-based** models for forecasting the sustainability of Open Source Software (OSS) projects (<https://github.com/OSS-PREY/OSSPREY-Pex-Forecaster>), achieving an 8% improvement in F1Score over SOTA

Software Engineer

July 2022 – August 2023

Center for Advanced Computing Research (CACR)

Dhaka, Bangladesh

- Developed the **AFMC Admission Test** Module, a software solution for the AFMC Entrance exam, utilizing **Node.js**, **Bootstrap**, and **MongoDB** to generate unique randomized questions for 30,000 candidates and automate the evaluation and ranking process
- Spearheaded a team to develop **C-Archive** using **PHP**, **MySQL**, and **Bootstrap** to store ongoing events of the department [[GITHUB](#)]

Patents

- Nafiz Imtiaz Khan**, Vladimir Filkov, "Systems And Methods For Artificial Intelligence-Driven Evidence-Based Action" (USPTO Filing ID: 32417/70825P) [[REF](#)]

Award & Honors

- **Champion** in the the **Medical Robotics Challenge for Contagious Diseases**, organized by **Imperial College London** [REF]
- **Champion** in the Creative App Contest of the **Tri Robo Cup Championship** organized by the **MIST Robotics club** [REF]
- **3rd runner-up** at **COVID-19 Combatants Unification Competition**, organized by **IEEE Bangladesh Chapter** [REF]
- **Top Downloaded Article Award** - in the year 2021, and 2022; provided by **Engineering Reports Journal - Wiley**
- **GGCS Summer Ph.D Fellowship** - 2024, awarded by Graduate Group of Computer Science, UC Davis
- Recognized for academic excellence at MIST, achieving placement on the **Dean's List of Honor** and receiving the **Merit Scholarship** in 2018, 2019, and 2020

Selected Publications

[Full List on Google Scholar]

1. **Nafiz Imtiaz Khan**, and Vladimir Filkov, "From Models to Practice: Enhancing OSS Project Sustainability with Evidence-Based Advice," in European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**ESEC/FSE-2024**). July 15-19, 2024, Porto de Galinhas, Brazil [LINK]
2. **Nafiz Imtiaz Khan**, and Vladimir Filkov, "EvidenceBot: A Privacy-Preserving, Customizable RAG-Based Tool for Enhancing Large Language Model Interactions" in European Software Engineering Conference and Symposium on the Foundations of Software Engineering (**ESEC/FSE-2025**). Mon 23 - Fri 27 June 2025 Trondheim, Norway [LINK]
3. **Nafiz Imtiaz Khan**, Priyal Soni, Arjun Ashok, and Vladimir Filkov, "OSSPREY: AI-Driven Forecasting and Intervention for OSS Project Sustainability" in IEEE/ACM International Conference on Automated Software Engineering (**ASE-2025**) Tool Demonstration Track. Sun 16 – Thu 20 November 2025, Seoul, South Korea [LINK]
4. **Nafiz Imtiaz Khan**, and Vladimir Filkov, Roger Eric Goldman, "AI-Driven Automation of Procedural Case Log Documentation" abstract presented at AMIA 2025 Clinical Informatics Conference [LINK]
5. **Nafiz Imtiaz Khan**, Vladimir Filkov, Kiley Cleland, Roger Eric Goldman, "Intelligent Documentation in Medical Education: Can AI Replace Manual Case Logging?" JAMIA Open [LINK]
6. Zeerak Babar, **Nafiz Imtiaz Khan**, Muhammad Hassanain, and Vladimir Filkov, "Open-Source LLMs for Technical Q&A: Lessons from StackExchange", International Conference on Software Engineering of Emerging Technologies (SEET), 2025. [LINK]
7. **Nafiz Imtiaz Khan**, and Vladimir Filkov, "Leveraging Language Models to Discover Evidence-Based Actions for OSS Sustainability" (Submitted) [LINK]
8. **Nafiz Imtiaz Khan**, Arjun Ashok, Swati Singhvi, Ștefan Stănculescu, and Vladimir Filkov, "Multi-Ecosystem Modeling of OSS Project Sustainability" (Submitted) [LINK]
9. **Nafiz Imtiaz Khan**, Tahasin Mahmud, Muhammad Nazrul Islam, and Sumaiya Nuha Mustafina, "Prediction of Cesarean Childbirth using Ensemble Machine Learning Methods", 22nd International Conference on Information Integration and Web-based Applications & Services (iiWAS2020), November 30-December 2, 2020, Chiang Mai, Thailand [LINK]
10. **Nafiz Imtiaz Khan**, Muhammad Nazrul Islam, and Tahasin Mahmud, "COVID-19 and Black Fungus: Analysis of Public Perceptions through Machine Learning", Engineering Reports. 2022 Apr;4(4):e12475. (**IF = 2 and SJR Rank = Q2**) [LINK]
11. Muhammad Nazrul Islam, Sumaiya Nuha Mustafina, Tahasin Mahmud, and **Nafiz Imtiaz Khan**, "Machine learning to predict pregnancy outcomes: a systematic review, synthesizing framework and future research agenda. BMC pregnancy and childbirth", 2022 Dec;22(1):1-9. (**IF = 3.09 and SJR Rank = Q1**) [LINK]
12. Jesika Rahman, Khondaker Sakil Ahmed, **Nafiz Imtiaz Khan**, Kamrul Islam, and Sujith Mangalathu, "Data-Driven Shear Strength Prediction of Steel Fiber Reinforced Concrete Beams Using Machine Learning Algorithms", Engineering Structures. 2021 Apr 15;233:111743.(**IF = 5.582 and SJR Rank = Q1**) [LINK]
13. Samia Zakir Sarothi, Khondaker Sakil Ahmed, **Nafiz Imtiaz Khan**, Aziz Ahmed, and Moncef Nehdi, "Predicting Bearing Capacity of Double Shear Bolted Connections Using Machine Learning", Engineering Structures, 2022 Jan 15;251:113497.(**IF = 5.582 and SJR Rank = Q1**) [LINK]
14. Mohammad Shahjahan Majib, Md. Mahbubur Rahman, T. M. Shahriar Sazzad, **Nafiz Imtiaz Khan**, and Samrat Kumar Dey, "VGG-SCNet: A VGG Net-based Deep Learning Framework for Brain Tumor Detection on MRI Images", IEEE Access, vol. 9, pp. 1880-1892, 2021. (**IF = 3.367 and SJR Rank = Q1**) [LINK]
15. Md Mahbubar Rahman, **Nafiz Imtiaz Khan**, Iqbal H. Sarker, Mohiuddin Ahmed, and Muhammad Nazrul Islam, "Leveraging machine learning to analyze sentiment from COVID-19 tweets: A global perspective", Engineering Reports. 2022:e12572. (**IF = 2 and SJR Rank = Q2**) [LINK]
16. Muhammad Nazrul Islam, **Nafiz Imtiaz Khan**, Ayon Roy, Md Mahbubar Rahman, Md Saddam Hossain Mukta, A.K.M Najmul Islam, "Sentiment Analysis of Bangladesh-specific COVID-19 Tweets using Deep Neural Network", 62nd International Conference on Information Technology and Management Science (ITMS'21), October 14-15, 2021, RIGA, LATVIA [LINK]